[INCH-POUND]
A−A−59789/3C
3 August 2020
SUPERSEDING
A−A−59789/3B
13 July 2015

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 5 PIECE, FOR COLD PLATE APPLICATIONS, .225 X .225 INCH BODY SIZE, WEDGE MOUNTING BODY, WITH TWO MOUNTING HOLES, WITH SCREW ACTUATION AND RETENTION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A−A−59789.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).



Example: AA59789/03NH60T is the PIN for a hard black anodize finished, 5.8 inch (147.3 mm) long card holder with a retained actuating screw. The card holder also feature two tapped mounting holes for use with 0−80 UNF 2B fasteners and a lockwasher and flat washer under the screw head for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card firmly in place providing high resistance to shock and vibration while providing maximum thermal transfer.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified herein and meet the general requirements specified in CID A−A−59789. Card holders shall have a screw retention feature to prevent unintentional disassembly of the card holder.

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A−A−59789.

Actuating screw hex drive socket. The dimension for hex drive socket shall be .094 inch (2.38 mm) across flats for mounting options "S", "R", "P", "U", and "T".

Cold plate slot width. The recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .250 inch (6.35 mm) plus the thickness of the printed board of the circuit card assembly (see A−A−59789).

Installation torque. The recommended nominal installation torque is as follows: 6 inch-pounds (0.68 N-m) for assemblies of configuration "A", "C", "F", or "W" and 7.2 inch-pounds (0.81 N-m) for assemblies of configurations "D", "J", "L", or "N".

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Configuration. The configuration of a card holder shall be as specified in table I. The details of a particular configuration consist of those on figures 1 and 2, and may include those on figures 3 and 4. This specification sheet covers two styles of mounting bodies, style A and style C. The differences between the two mounting body styles are detailed in tables I and II and on figure 1.

TABLE I. Configurations.

Configuration	Mounting body style	Finish options	Mounting options	Applicable figures	Hardware options
A	A	B, C, H, P	S, R, T, P	1 and 2	No hardware options added
C	C	B, C, E, H	S, R, T, U		
F	A	B, C, H, P	S, R, T, P	1, 2, and 3	Lockwasher and flat washer
W	C	B, C, E, H	S, R, T, U		
N	A	B, C, H, P	S, R, T, P	1, 2, and 4	Screw self-lock element
L	C	B, C, E, H	S, R, T, U		
J	A	B, C, H, P	S, R, T, P	1, 2, 3, and 4	Lockwasher, flat washer, and screw self-lock element
D	C	B, C, E, H	S, R, T, U		

Finish. The finish designator shall be as specified in A−A−59789. The finishes available for this CID specification sheet are as follows: "B" (black anodize), "C" (gold chemical film), "E" (electroless nickel), "P" (electrodeposited nickel), or "H" (hard black anodize). See table I for the finish options associated with the two mounting body styles. The center mounting wedge is not finished for card holders using mounting option "P".

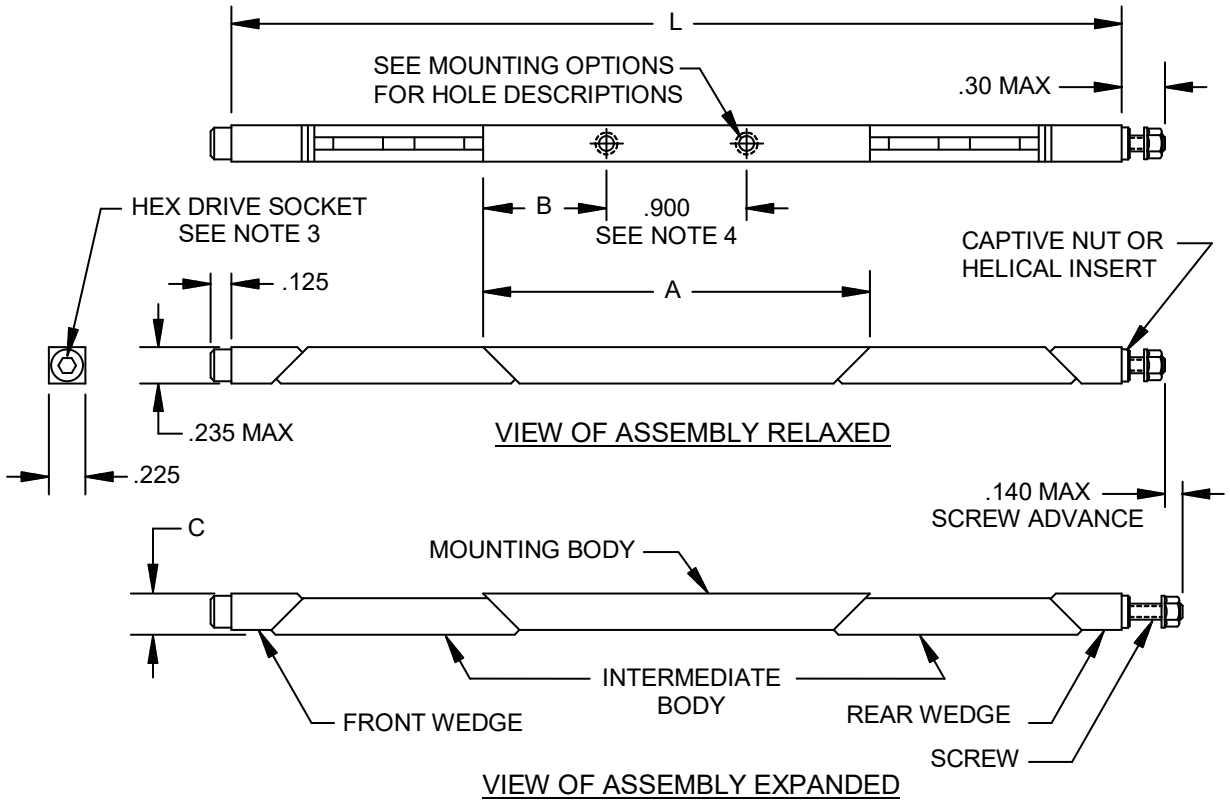
Length, expanded and relaxed dimensions. The length designator shall be as specified in A−A−59789 and the lengths available for this CID specification sheet are listed in table II. The length, expanded, and relaxed dimensions shall be as specified on figure 1.

TABLE II. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension "L" ±.02 (0.5)	Mounting body style	Dimension "A" ±.01 (0.3)	Dimension "B" ±.02 (0.5)	Dimension "C" minimum
50	4.8 (121.9)	A	1.50 (38.1)	.30 (7.6)	.275 (6.99)
		C	1.50 (38.1)	.30 (7.6)	.275 (6.99)
55	5.3 (134.6)	A	2.00 (50.8)	.55 (14.0)	.275 (6.99)
		C	1.50 (38.1)	.30 (7.6)	.275 (6.99)
60	5.8 (147.3)	A	2.50 (63.5)	.80 (20.3)	.275 (6.99)
		C	1.50 (38.1)	.30 (7.6)	.275 (6.99)
65	6.3 (160.0)	A	3.00 (76.2)	1.05 (26.7)	.275 (6.99)
		C	1.50 (38.1)	.30 (7.6)	.275 (6.99)

1/ Dimensions are in inches. Millimeters are given for information only.

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Inches	mm	Inches	mm	Inches	mm
.125	3.18	.225	5.72	.30	7.6
.140	3.56	.235	5.97	.900	22.86

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 (0.25 mm) for three place decimals.
3. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm).
4. Tolerance for the hole spacing is ±.010 inch (0.25 mm) for mounting body style A and ±.005 inch (0.013 mm) for mounting body style C.

FIGURE 1. Relaxed and expanded dimensions.

Mounting. The mounting designators shall be as specified in A−A−59789. The mounting options available for this CID specification sheet are as follows: "S" (tapped 2-56 holes), "R" (rivet mount holes with counterbore and countersink), "T" (tapped 0-80 holes), "P" (Ø.062 x .040 index pins), or "U" (tapped metric M2 x 0.4 holes). See figure 1 and table II for mounting hole spacing requirements. See table I for the mounting options associated with the two mounting body styles. Card holders using mounting option "P" and "R" may be shipped unassembled.

Rivet mounting holes. The holes used for rivet mounting shall be .066/.073 inch (1.68/1.85 mm) diameter, countersunk 100 degrees by .140 inch (3.56 mm) diameter through holes with a counterbore of .156 inch (3.96 mm) diameter with a depth of .165 inch (4.19 mm) for style C mounting bodies and .175 inch (4.45 mm) for style A mounting bodies.

Rivets. This card holder uses rivet type A as specified in A−A−59789 when rivet mounting is used.

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Screw retention locknut (see figure 2). Card holders shall have a screw retention locknut intended to prevent unintentional disassembly of the card holder. The locknut shall be fabricated of a corrosion resistant material.

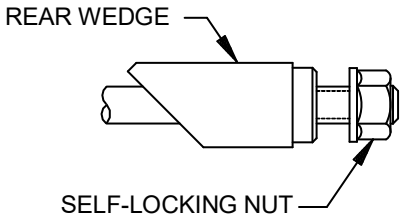


FIGURE 2. Screw retaining self-locking nut (corrosion resistant) details.

Configuration hardware options. Card holders can have the following options: No hardware options added, lockwasher and flat washer (see figure 3), screw self lock element (see figure 4), or lockwasher, flat washer, and screw self lock element (see figures 3 and 4). See table I for the correct PIN configuration identifier.

No hardware options added. The base configuration of the card holder is as depicted by figures 1 and 2. Card holders not requiring any hardware options shall include configuration identifier "A" or "C" in the PIN (see classification, notes, and table I).

Lockwasher and flat washer (see figure 3). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer shall include configuration identifier "F" or "W" in the PIN (see classification, notes, and table I).

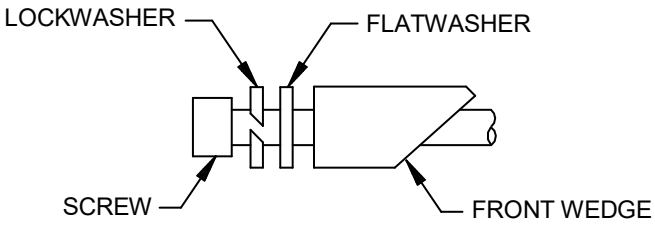


FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock vibration. Card holders requiring a screw self-locking element shall include configuration identifier "L" or "N" in the PIN (see classification, notes, and table I).

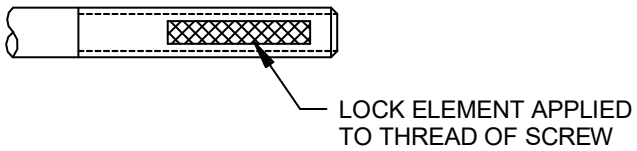


FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include configuration identifier "D" or "J" in the PIN (see classification, notes, and table I).

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NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A−A−59789 − Holder, Electrical Card, Wedge Retainers, 5 Piece, For Cold Plate Applications, General Requirements For.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

Ordering data. Ordering data shall be as specified in A−A−59789.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
5BG68	American Circuit Card Retainers, Inc. 2310 E. Orangethorpe Avenue Anaheim, CA 92806−1231	Telephone: (714) 738−6194 Facsimile: (714) 446−0119 E-mail: sales@accrmfg.com URL: https://www.accrmfg.com
61081 (1)	Schroff, Inc. (formerly Birtcher) 7328 Trade Street San Diego, CA 92121−3410	Telephone: (858) 740−2400 Facsimile: (858) 740−2430 E-mail: askschroff@nVent.com URL: https://schroff.nvent.com
61081 (2)	Schroff, Inc. (formerly Calmark) 7328 Trade Street San Diego, CA 92121−3410	Telephone: (858) 740−2400 Facsimile: (858) 740−2430 E-mail: askschroff@nVent.com URL: https://schroff.nvent.com

Part number supersession data. This CID specification sheet PINs supersedes the following manufacturer's part numbers as shown in tables III and IV. The CID PINs listed in tables III and IV are only for length designator "60". See table V for CID PIN construction using other available lengths for this CID specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

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TABLE III. Commercial part number supersession data, mounting body style A.

PIN designator AA59789/03	Vendor similar designator or type part number 1/ CAGE 61081 2/	PIN designator AA59789/03	Vendor similar designator or type part number 1/ CAGE 61081 2/
AB60S	42-5-12-B-LN	NB60S	42-5-12-B-LN-L
AB60R	42-5-12-B-LN-R	NB60R	42-5-12-B-LNRL
AB60T	42-5-12-B-LN-T	NB60T	42-5-12-B-LNTL
AB60P	42-5-12-B-LN-P	NB60P	42-5-12-B-LNPL
FB60S	42-5-12-B-LFLN	JB60S	42-5-12-BLFLNL
FB60R	42-5-12-BLFLNR	JB60R	42-5-12BLFLNR
FB60T	42-5-12-BLFLNT	JB60T	42-5-12BLFLNTL
FB60P	42-5-12-BLFLNP	JB60P	42-5-12BLFLNPL
AC60S	42-5-12-LN	NC60S	42-5-12-LN-L
AC60R	42-5-12-LN-R	NC60R	42-5-12-LN-R-L
AC60T	42-5-12-LN-T	NC60T	42-5-12-LN-T-L
AC60P	42-5-12-LN-P	NC60P	42-5-12-LN-P-L
FC60S	42-5-12-LF-LN	JC60S	42-5-12-LF-LNL
FC60R	42-5-12-LF-LNR	JC60R	42-5-12-LFLNRL
FC60T	42-5-12-LF-LNT	JC60T	42-5-12-LFLNTL
FC60P	42-5-12-LF-LNP	JC60P	42-5-12-LFLNPL
AP60S	42-5-12-N-LN	NP60S	42-5-12-N-LN-L
AP60R	42-5-12-N-LN-R	NP60R	42-5-12-N-LNRL
AP60T	42-5-12-N-LN-T	NP60T	42-5-12-N-LNTL
AP60P	42-5-12-N-LN-P	NP60P	42-5-12-N-LNPL
FP60S	42-5-12-N-LFLN	JP60S	42-5-12-NLFLNL
FP60R	42-5-12-NLFLNR	JP60R	42-5-12NLFLNRL
FP60T	42-5-12-NLFLNT	JP60T	42-5-12NLFLNTL
FP60P	42-5-12-NLFLNP	JP60P	42-5-12NLFLNPL

See footnotes at end of table.

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TABLE III. Commercial part number supersession data, mounting body style A - Continued.

PIN designator AA59789/03	Vendor similar designator or type part number 1/ CAGE 61081 2/	PIN designator AA59789/03	Vendor similar designator or type part number 1/ CAGE 61081 2/
AH60S	42-5-12-B3-LN	NH60S	42-5-12-B3-LNL
AH60R	42-5-12-B3-LNR	NH60R	42-5-12-B3LNRL
AH60T	42-5-12-B3-LNT	NH60T	42-5-12-B3LNTL
AH60P	42-5-12-B3-LNP	NH60P	42-5-12-B3LNPL
FH60S	42-5-12-B3LFLN	JH60S	42-5-12B3LFLNL
FH60R	42-5-12B3LFLNR	JH60R	42-512B3LFLNRL
FH60T	42-5-12B3LFLNT	JH60T	42-512B3LFLNTL
FH60P	42-5-12B3LFLNP	JH60P	42-512B3LFLNPL

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A-A-59789.

2/ CAGE is for 61081 (1).

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TABLE IV. Commercial part number supersession data, mounting body style C.

PIN designator AA59789/03	Vendor similar designator or type part number 1/		
	CAGE 61081 2/	CAGE 5BG68	CAGE
CB60S	A231CR5.80T2R	G5220BA-5.80T2R	
CB60R	A231CR5.80HR	GG5220BA-5.80HR	
CB60T	A231CR5.80T0R	G52220BA-5.80T0R	
CB60U	A231CR5.80TM2R	G5220BA-5.80TM2R	
WB60S	WA231CR5.80T2R	G5220WBA-5.80T2R	
WB60R	WA231CR5.80HR	G5220WBA-5.80HR	
WB60T	WA231CR5.80T0R	G5220WBA-5.80T0R	
WB60U	WA231CR5.80TM2R	G5220WBA-5.80TM2R	
LB60S	A231CR5.80T2LR	G5220BA-5.80T2LR	
LB60R	A231CR5.80HLR	G5220BA-5.80HLR	
LB60T	A231CR5.80T0LR	G5220BA-5.80T0LR	
LB60U	A231CR5.80TM2LR	G5220BA-5.80TM2LR	
DB60S	WA231CR5.80T2LR	G5220WBA-5.80T2LR	
DB60R	WA231CR5.80HLR	G5220WBA-5.80HLR	
DB60T	WA231CR5.80T0LR	G5220WBA-5.80T0LR	
DB60U	WA231CR5.80TM2LR	G5220WBA-5.80TM2LR	
CC60S	231CR5.80T2R	G5220CG-5.80T2R	
CC60R	231CR5.80HR	G5220CG-5.80HR	
CC60T	231CR5.80T0R	G5220CG-5.80T0R	
CC60U	231CR5.80TM2R	G5220CG-5.80TM2R	
WC60S	W231CR5.80T2R	G5220WCG-5.80T2R	
WC60R	W231CR5.80HR	G5220WCG-5.80HR	
WC60T	W231CR5.80T0R	G5220WCG-5.80T0R	
WC60U	W231CR5.80TM2R	G5220WCG-5.80TM2R	

See footnotes at end of table.

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TABLE IV. Commercial part number supersession data, mounting body style C - Continued.

PIN designator AA59789/03	Vendor similar designator or type part number 1/		
	CAGE 61081 2/	CAGE 5BG68	CAGE
LC60S	231CR5.80T2LR	G5220CG-5.80T2LR	
LC60R	231CR5.80HLR	G5220CG-5.80HLR	
LC60T	231CR5.80T0LR	G5220CG-5.80T0LR	
LC60U	231CR5.80TM2.5LR	G5220CG-5.80TM2LR	
DC60S	W231CR5.80T2LR	G5220WCG-5.80T2LR	
DC60R	W231CR5.80HLR	G5220WCG-5.80HLR	
DC60T	W231CR5.80T0LR	G5220WCG-5.80T0LR	
DC60U	W231CR5.80TM2.5LR	G5220WCG-5.80TM2LR	
CE60S	EN231CR5.80T2R	G5220EN-5.80T2R	
CE60R	EN231CR5.80HR	G5220EN-5.80HR	
CE60T	EN231CR5.80T0R	G5220EN-5.80T0R	
CE60U	EN231CR5.80TM2R	G5220EN-5.80TM2R	
WE60S	WEN231CR5.80T2R	G5220WEN-5.80T2R	
WE60R	WEN231CR5.80HR	G5220WEN-5.80HR	
WE60T	WEN231CR5.80T0R	G5220WEN-5.80T0R	
WE60U	WEN231CR5.80TM2R	G5220WEN-5.80TM2R	
LE60S	EN231CR5.80T2LR	G5220EN-5.80T2LR	
LE60R	EN231CR5.80HLR	G5220EN-5.80HLR	
LE60T	EN231CR5.80T0LR	G5220EN-5.80T0LR	
LE60U	EN231CR5.80TM2LR	G5220EN-5.80TM2LR	
DE60S	WEN231CR5.80T2LR	G5220WEN-5.80T2LR	
DE60R	WEN231CR5.80HLR	G5220WEN-5.80HLR	
DE60T	WEN231CR5.80T0LR	G5220WEN-5.80T0LR	
DE60U	WEN231CR5.80TM2LR	G5220WEN-5.80TM2LR	

See footnotes at end of table.

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TABLE IV. Commercial part number supersession data, mounting body style C - Continued.

PIN designator AA59789/03	Vendor similar designator or type part number 1/		
	CAGE 61081 2/	CAGE 5BG68	CAGE
CH60S	HA231CR5.80T2R	G5220BH-5.80T2R	
CH60R	HA231CR5.80HR	G5220BH-5.80HR	
CH60T	HA231CR5.80T0R	G5213BH-5.80T0R	
CH60U	HA231CR5.80TM2R	G5220BH-5.80TM2R	
WH60S	WHA231CR5.80T2R	G5220WBH-5.80T2R	
WH60R	WHA231CR5.80HR	G5220WBH-5.80HR	
WH60T	WHA231CR5.80T0R	G5220WBH-5.80T0R	
WH60U	WHA231CR5.80TM2R	G5220WBH-5.80TM2R	
LH60S	HA231CR5.80T2LR	G5220BH-5.80T2LR	
LH60R	HA231CR5.80HLR	G5220BH-5.80HLR	
LH60T	HA231CR5.80T0LR	G5220BH-5.80T0LR	
LH60U	HA231CR5.80TM2LR	G5220BH-5.80TM2LR	
DH60S	WHA231CR5.80T2LR	G5220WBH-5.80T2LR	
DH60R	WHA231CR5.80HLR	G5220WBH-5.80HLR	
DH60T	WHA231CR5.80T0LR	G5220WBH-5.80T0LR	
DH60U	WHA231CR5.80TM2LR	G5220WBH-5.80TM2LR	

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A-A-59789.

2/ CAGE is for 61081 (2).

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PIN length examples. The CID PINs listed in table V are for all available standard card holder lengths for this specification sheet. However, only one specific finish, mounting and configuration are listed (see PIN example for a break-down of the codes).

TABLE V. Example of PIN with available length designators.

PIN designator AA59789/03	Vendor similar designator or type part number 1/ 2/			
	CAGE 61081 3/	CAGE 61081 4/	CAGE 5BG68	CAGE
NH50T	42−5−10−B3LNTL			
NH55T	42−5−11−B3LNTL			
NH60T	42−5−12−B3LNTL			
NH65T	42−5−13−B3LNTL			
LH50T		HA231CR4.80T0LR	G5220BH−4.80T0LR	
LH55T		HA231CR5.30T0LR	G5220BH−5.30T0LR	
LH60T		HA231CR5.80T0LR	G5220BH−5.80T0LR	
LH65T		HA231CR6.30T0LR	G5220BH−6.30T0LR	

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A−A−59789.
2/ Other lengths are available on request.
3/ CAGE is for 61081 (1).
4/ CAGE is for 61081 (2).

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

MILITARY INTERESTS:

Custodians:
Army − CR
Navy − EC
Air Force − 85
DLA − CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA − FAS

Preparing Activity:
DLA − CC

Project 5998−2020−023

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